

Pankaj Kumar

Senior Data Engineer

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Professional Summary

Databricks-certified Senior Data Engineer with nearly 6 years building Lakehouse platforms and streaming pipelines on Azure Databricks and GCP. Work spans 10M+ daily events at sub-5-second latency, 30% faster processing, and regulated data across healthcare, banking, and telecom.

Technical Skills

Programming: Python, SQL, PySpark, Apache Spark

Databricks & Streaming: Databricks (Unity Catalog, Delta Lake, Workflows, Asset Bundles), Spark Structured Streaming, Kafka

Cloud & Platforms: Azure (Databricks, Data Factory, ADLS), GCP (BigQuery, Dataproc, Composer), Snowflake

Architecture & Modeling: Lakehouse, Medallion Architecture, Dimensional Modeling, SCD Type 1/2, ETL/ELT Pipelines

Governance & Quality: Data Quality Frameworks, Lineage, PII & Access Controls, RBAC, Data Governance

Tools & DevOps: Apache Airflow, dbt, GitLab, Jenkins, Azure DevOps, Terraform, Unix Shell, CI/CD

Professional Experience

Senior Data Engineer

Apr 2025 – Present

Harman India (HCS) | Projects: Convatec (Patient Platform), Lowe's (Price Optimisation & Strategy)

- Architected a config-driven Bronze to Silver data-quality engine on **Azure Databricks** (Unity Catalog, Delta Lake, Workflows) for a healthcare RWE Lakehouse, pairing automated rule-based validation with incremental CDC/Delta MERGE. New clinical sources onboard through JSON config alone, reusing **90%** of the code.
- Environment promotion and release automation for every data workflow run through **Databricks Asset Bundles**, **Terraform**-provisioned Azure infrastructure, and Azure DevOps across dev, test, and prod.
- Designed and built a production **Snowflake** pipeline with **dbt** and Snowpark, orchestrated via Airflow, securing healthcare data with RBAC, dynamic data masking, and row-access policies.
- Streaming pipelines on **Kafka** and Spark Structured Streaming carry **10M+ daily events** at sub-5-second latency, keeping GCP and on-premises systems in near-real-time sync.
- Migrated legacy Hadoop/Oozie workloads to **Cloud Composer (Airflow)** and tuned the Spark jobs behind them (partitioning, caching, OPTIMIZE/Z-ORDER); pipeline failures dropped **20%**.
- Engineered a historical data warehouse in **BigQuery** with dimensional modeling, then added Dataplex quality checks and Looker dashboards that took incident mitigation time down **30%**.

Data Engineer

May 2023 – Apr 2025

Ernst & Young Private Limited (Client: HSBC Bank)

- Built a metadata-driven model execution framework in **Python**, with variable derivation, calculation order, and source/target mappings held as config in BigQuery. Financial model migration time fell **20%**.
- Migrated SAS-based financial models to GCP Dataproc using **PySpark**; processing times fell **30%** and the legacy dependencies were retired.
- Developed modular, version-controlled **dbt** models on BigQuery (staging to marts) with incremental materializations and automated dbt tests, which cut manual validation effort **25%** and kept lineage audit-ready for regulated reporting.
- Reconciliation scripts in **Python** validate every migrated model against its legacy SAS baseline, holding data accuracy at **97%** across regulated financial reporting.

Data Engineer

Oct 2020 – May 2023

Capgemini Technology Services India Ltd. (Client: Rogers Communications)

- Ingested SAS, Oracle, and Hive sources into Azure Data Lake through orchestrated PySpark and **Azure Data Factory** pipelines, scaling to over 3 million daily incremental records with SCD Type 1/2 for **10M+ subscribers**.
- Built subscriber churn analytics on Azure Databricks with **medallion architecture** (bronze/silver/gold), processing millions of daily CDRs to surface at-risk customers.
- Proactive monitoring and Spark ETL tuning on subscriber workloads brought pipeline downtime down **20%**, MTTR **15%**, and the job failure rate **10%**.

Certifications

Databricks Certified Data Engineer Associate

2026

Education

Bachelor of Engineering in Computer Science (Big Data Analytics by IBM)

Chandigarh University | 2016 – 2020